Lab 02 - Detect Actionable Service Signals in R

Course: DevOps Foundation Training (TGS-2024049184) Version: v4.0 Topic: 02 Estimated time: 35-45 minutes

Workplace scenario

Checkout latency has intermittent spikes. The operations channel receives too many generic alerts, so the team needs an evidence-based detection rule and clear routing fields.

Objective

Apply K2 and A2 by designing an automated detection rule that produces specific, severity-ranked problem signals without excessive noise.

What you will produce

A detected_alerts.csv with severity, evidence and action plus a telemetry_detection.png chart.

Prerequisites

- R 4.x available as Rscript.
- Work from this lab folder so relative paths resolve.
- Use only the supplied synthetic data; no internet connection or external package is required.
- Preserve the source data. Make experimental changes in a copied file, then restore it.

Folder map

```
Lab-02-automated-signal-detection/
■■■ README.md / README.pdf
■■■ CHECKLIST.md / CHECKLIST.pdf
■■■ automated_signal_detection.R
■■■ data/
■■■ output/            # reproducible evidence produced by the R script
```

Detailed procedure

Step 1 - Review the telemetry sample and identify the service-level indicators represented by each column.

```
Rscript -e 'd <- read.csv("data/telemetry.csv"); summary(d)'
```

Record what you observed before continuing.

Step 2 - Run automated_detection.R to calculate a rolling baseline and flag only sustained abnormal windows.

```
Rscript automated_detection.R
```

Record what you observed before continuing.

Step 3 - Inspect detected_alerts.csv; confirm every alert names the service, signal, severity, evidence and immediate action.

```
Rscript -e 'print(read.csv("output/detected_alerts.csv"))'
```

Record what you observed before continuing.

Step 4 - Open the detection chart and compare the threshold with the observed latency series.

```
open output/telemetry_detection.png
```

Record what you observed before continuing.

Step 5 - Change min_consecutive from 2 to 1, rerun, and compare the alert count to demonstrate alert-fatigue risk; then restore it to 2.

```
Rscript automated_detection.R
```

Record what you observed before continuing.

Step 6 - Complete CHECKLIST.md with the chosen rule, trade-off and escalation owner.

```
open CHECKLIST.md
```

Record what you observed before continuing.

Evidence to retain

[ ] output/detected_alerts.csv

[ ] output/telemetry_detection.png

[ ] completed CHECKLIST.md

Acceptance criteria

- Two sustained alert windows
- No alert for isolated row 7 spike
- Every alert has severity, owner and action
- Learner explains sensitivity versus noise

Verification

The final rule emits exactly two sustained alert windows, no single-point spike alert, and every alert has a non-empty owner and action.

Troubleshooting

- cannot open file: confirm your terminal is in this lab folder.
- Rscript: command not found: install R, reopen the terminal and run Rscript --version.

- A validation stop(...) message is evidence that the input broke a stated rule; inspect the named field rather than bypassing the check.
- If an output looks wrong, compare row counts and IDs with the source CSV before changing the analysis.

Clean rerun

Delete only the files inside output/, then run the R script again. The source CSV and script remain unchanged.